

Course Content

Introduction To Automation

Development Delivery Pipeline Overview, Automating the build pipeline, RAD (Rapid Application Development), Code generation, MDA/MDD (Model-Driven Architecture/Development)

UNIT II: 4 Lecture Hours **Advantages of Automation**

Scenarios where automation saves time and effort, Scenarios where automation prevent errors.

UNIT III: 4 Lecture Hours **Interacting With Linux Environment**

Understanding Linux file system, User groups and permissions, Working with Bash

UNIT IV: 4 Lecture Hours **Scripting Development Tasks** Writing automation scripts, Best practices for scripting

UNIT V: 3 Lecture Hours Understanding Make And Makefiles

UNIT VI 3Lecture Hours **Role of Administrator** Different from a developer, Critical tasks, Installing Linux

UNIT VII 3Lecture Hours **Delving Deep Into Filesystem Hierarchy**

Understanding hierarchy, root and usr directory, GRUB and MBR

UNIT VIII 3Lecture Hours **Package Management Systems**

Yum, apt

UNIT IX 3Lecture Hours **Configuration And Maintenance**

Managing public and private groups, Linux groups, Adding users to groups, Evaluating private group usage

UNIT X 3Lecture Hours **Monitoring In Linux** Metrics to monitor, In built Linux monitoring tools

UNIT XI 3Lecture Hours **Need of Third Party Tools For Monitoring**

Text Books

1. Advanced Linux – Volume 1, Xebia Press
2. Development Automation – Volume 1, Xebia Press

Reference Books:

1. Running Linux – Book by Matthias Kalle Dalheimer, Matt Welsh
2. Linux System Administration - Tom Adelstein, Bill Lubanovic
3. Mastering Linux Shell Scripting – Book by Andrew Mallett